

Day 18 - Lecture 18.18: Advanced Customizations on Pie Charts

1. Overview & Visual Emphasis Techniques

Standard flat pie charts can look plain. Matplotlib allows rich customizations to **emphasize specific categories**, improve aesthetics, and control borders:

- `explode` : Pulls a specific slice outward from the pie center.
 - `shadow=True` : Adds depth and drop-shadow styling.
 - `startangle` : Rotates the pie so the primary slice begins at a specific angle (e.g. 90° top or 180°).
 - `wedgeprops` : Controls border width, edgecolor, and fill transparency.
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2. Advanced Parameters & Wedgeprops

```
wedgeprops = {  
    'edgecolor': 'black',      # Border outline color  
    'linewidth': 1.5,         # Outline thickness  
    'linestyle': '--',        # Dashed border  
    'fill': True               # Whether to fill slice  
}
```

Explode Offset Vector:

An array of floats of equal length to categories, where each value represents the fractional radius offset:

```
# Pulls only the 4th slice ("R&D") outward by 20% of radius:  
explode = [0, 0, 0, 0.2, 0]
```

3. Code Implementation

```
import matplotlib.pyplot as plt

expenses = ["Salaries", "Rent", "Marketing", "R&D", "Miscellaneous"]
amounts = [500, 150, 120, 100, 50]

# Explode the 4th category ("R&D")
explode = [0, 0, 0, 0.15, 0]
colors = ["#4e79a7", "#f28e2b", "#e15759", "#76b7b2", "#59a14f"]

plt.figure(figsize=(8, 8))
plt.pie(
    amounts,
    labels=expenses,
    autopct='%1.1f%%',
    explode=explode,
    shadow=True,
    startangle=90,          # Starts highest slice at 12 o'clock position
    colors=colors,
    wedgeprops={"edgecolor": "black", "linewidth": 1.2}
)

plt.title("Budget Distribution with Exploded Slice (R&D) & Shadow", fontsize=14,
plt.tight_layout()
plt.show()

# Donut Chart Concept using wedgeprops
plt.figure(figsize=(7, 7))
plt.pie(
    amounts,
    labels=expenses,
    autopct='%1.1f%%',
    pctdistance=0.75,
    colors=colors,
    wedgeprops=dict(width=0.4, edgecolor='white', linewidth=2) # width < 1 creat
)
plt.title("Modern Donut Chart (Budget Breakdown)", fontsize=14, fontweight="bold"
plt.tight_layout()
plt.show()
```

4. Key Takeaways

- Setting `wedgeprops=dict(width=0.4)` automatically turns a pie chart into a modern **Donut Chart**, which is widely preferred in corporate dashboards because the empty center can display the total budget (\$920K).
- Use `startangle=90` so the largest slice starts at 12 o'clock and rotates clockwise.